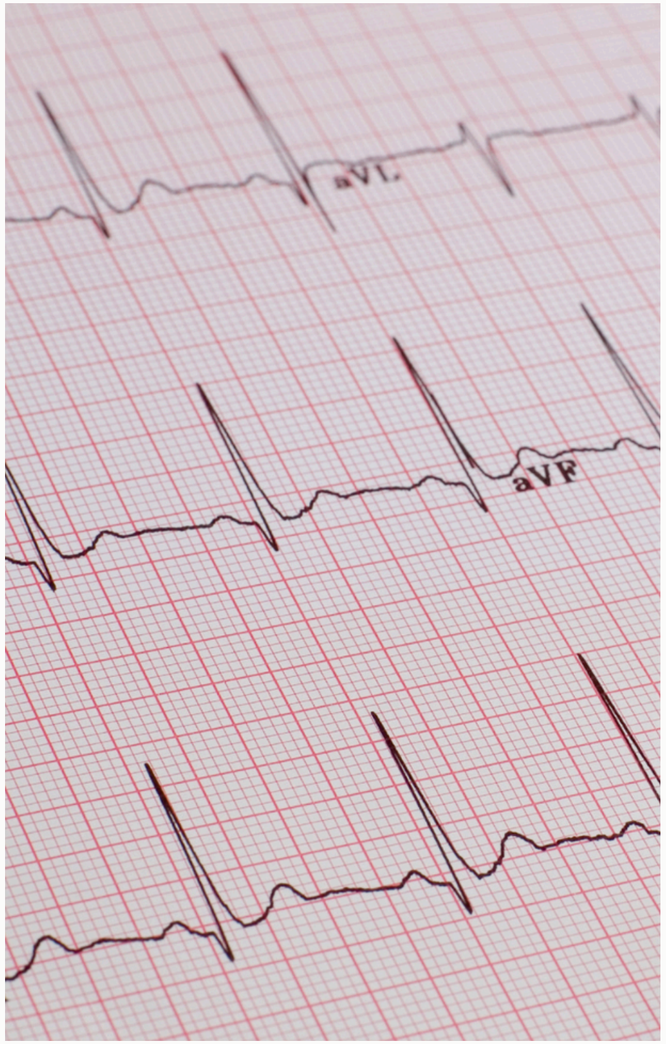
BM4152 BIOSIGNAL PROCESSING · PAPER IMPLEMENTATION · INITIAL PRESENTATION

Combining Low-Dimensional Wavelet Features and Support Vector Machine for Arrhythmia Beat Classification

Qin et al., Scientific Reports (2017)

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SECTION 01

Introduction & Background

The Problem & The Gap

- An arrhythmia is an irregular, too fast, or too slow heartbeat caused by a problem with the heart's electrical signals.
- ECG-based arrhythmia detection is essential, but manual review by clinicians is slow and doesn't scale for long-term monitoring
- Traditional classification pipeline: feature extraction → feature selection → classification
- This paper's contribution: a low-dimensional wavelet + PCA feature pipeline, evaluated under both beat-based and record-based (patient-independent) 10-fold cross-validation
- Most prior work reports beat-based accuracy, where beats from the same patient can appear in both training and test sets - this inflates real-world performance

Source: Qin, Q. et al. Sci Rep 7, 6067 (2017)

SECTION 02

Dataset

MIT-BIH Arrhythmia Database

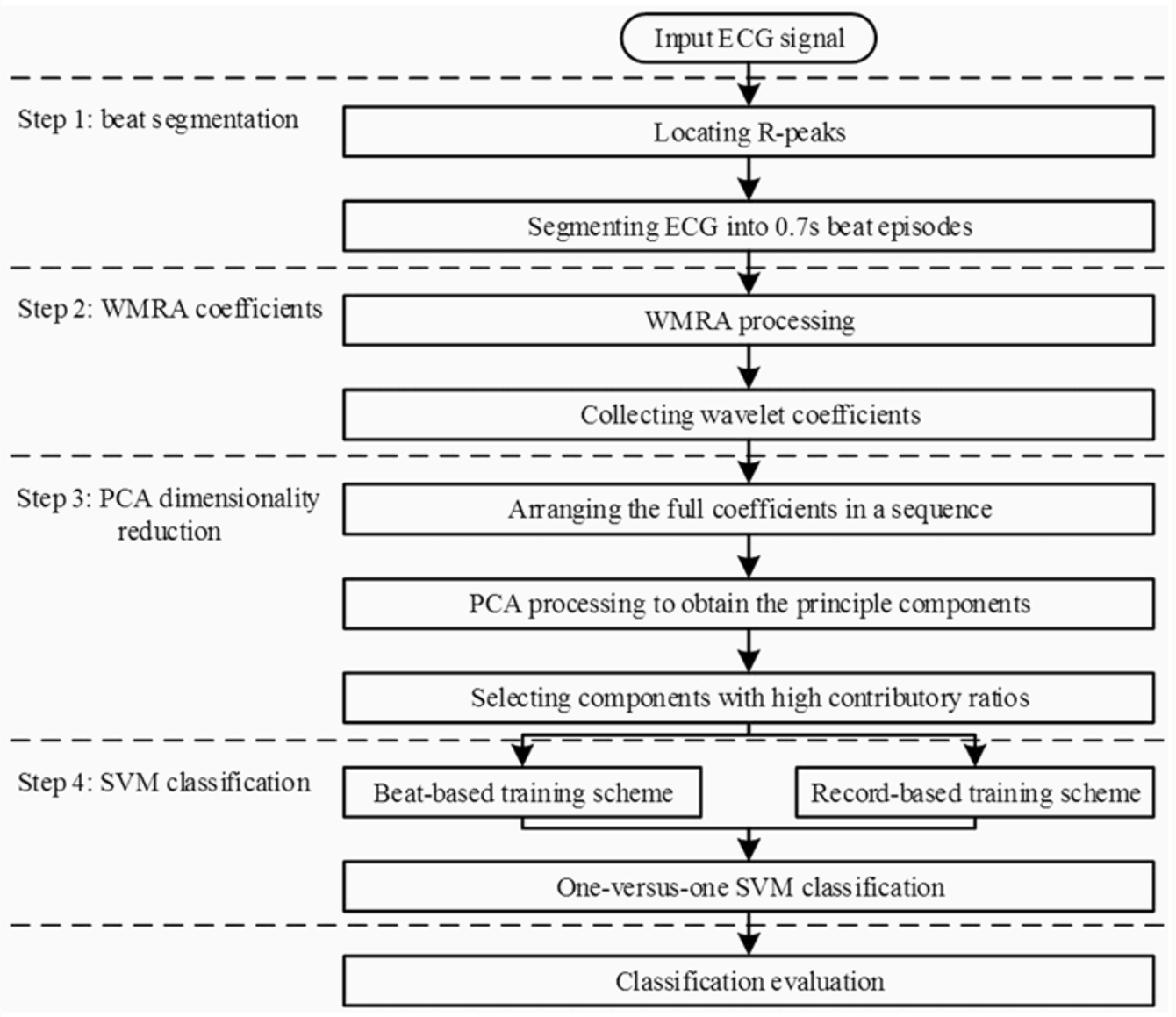
Attribute	Value
Records	48 ECG records, Lead II, 30 minutes each
Sampling	360 Hz, 11-bit resolution
Beat types used	A, L, N, P, R, V (6 of 16 annotated arrythmia types)
Total beats used	107,049 of 109,966 total beats
Class balance	Highly imbalanced - Normal (N) beats are ~70% of the data

Source: Moody & Mark, MIT-BIH Arrhythmia Database (PhysioNet)

SECTION 03

Methods

Four-Step Pipeline



Adapted from Qin et al. (2017), Fig. 1

Steps 1 : Segmentation

- Each beat is cut into a 252-sample (0.7 s) window: 0.25 s before the R-peak + 0.45 s after, covering the P wave, QRS complex, and T wave

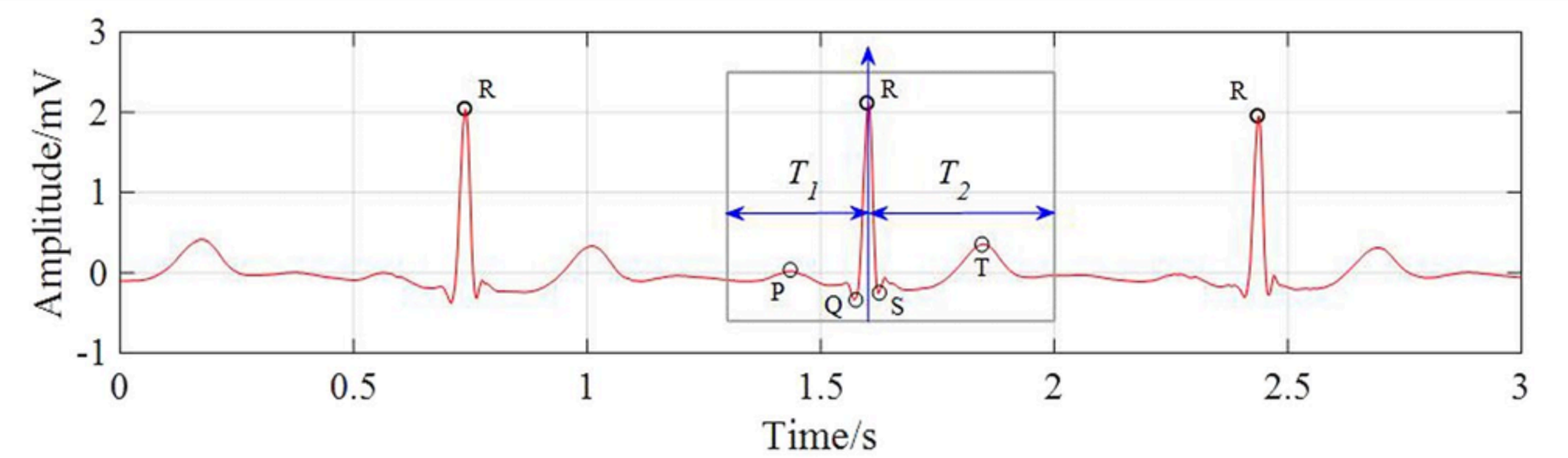


Figure 2. Beat segmentation of a typical ECG signal.

Source: Qin, Q. et al. Sci Rep 7, 6067 (2017)

Steps 2 : Wavelet Features

- Each window is decomposed with an 8-level Wavelet Multi-Resolution Analysis using the bior6.8 wavelet
- Detail coefficients cD1–cD8 (≈ 0.70–180 Hz) capture the informative ECG band and are kept
- Approximation coefficient cA8 (0–0.70 Hz, mostly baseline drift) is discarded
- Result: a 373-element raw wavelet feature vector per beat

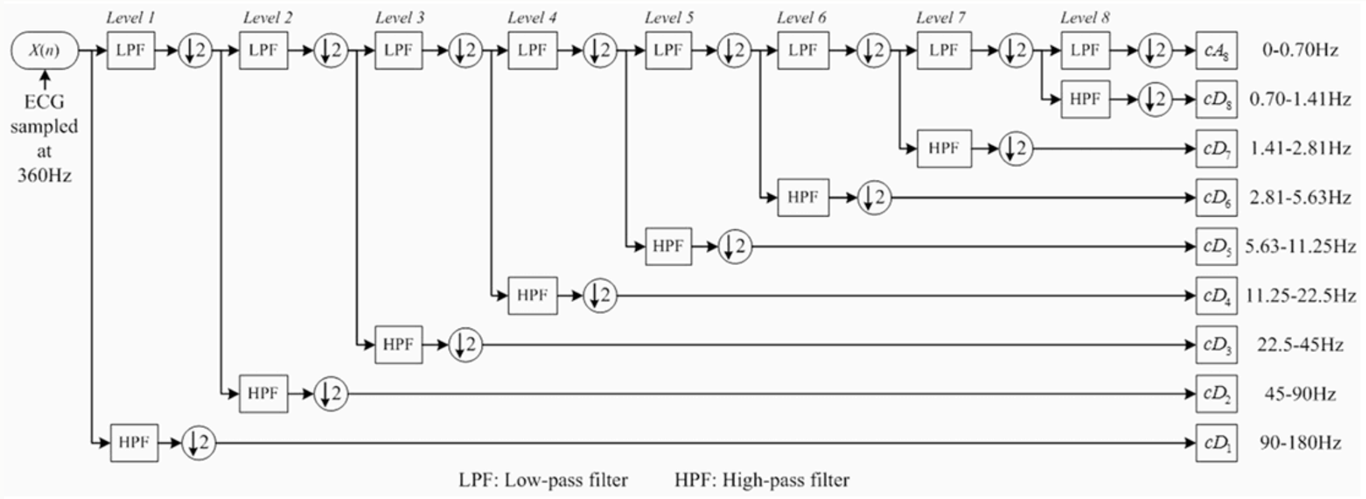


Figure 3. The decomposition process of the 8-level WMRA.

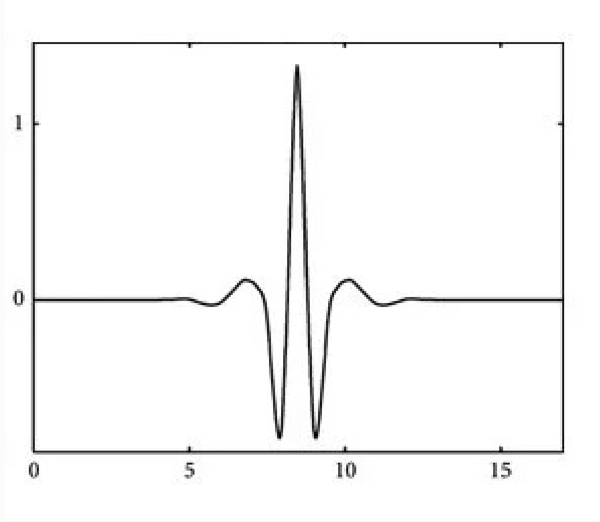


Figure 4. Mother wavelet function Bior6.8 for decomposition

Source: Qin, Q. et al. Sci Rep 7, 6067 (2017)

Steps 3: PCA Reduction

- We have a 373-element vector per every heartbeat - these can contain redundant,correlated information
- PCA is applied directly to the flattened 373-element vector to reduce the dimentionality
- Accuracy plateaus at 12 principal components (~97% dimensionality reduction) for both cross-validation schemes

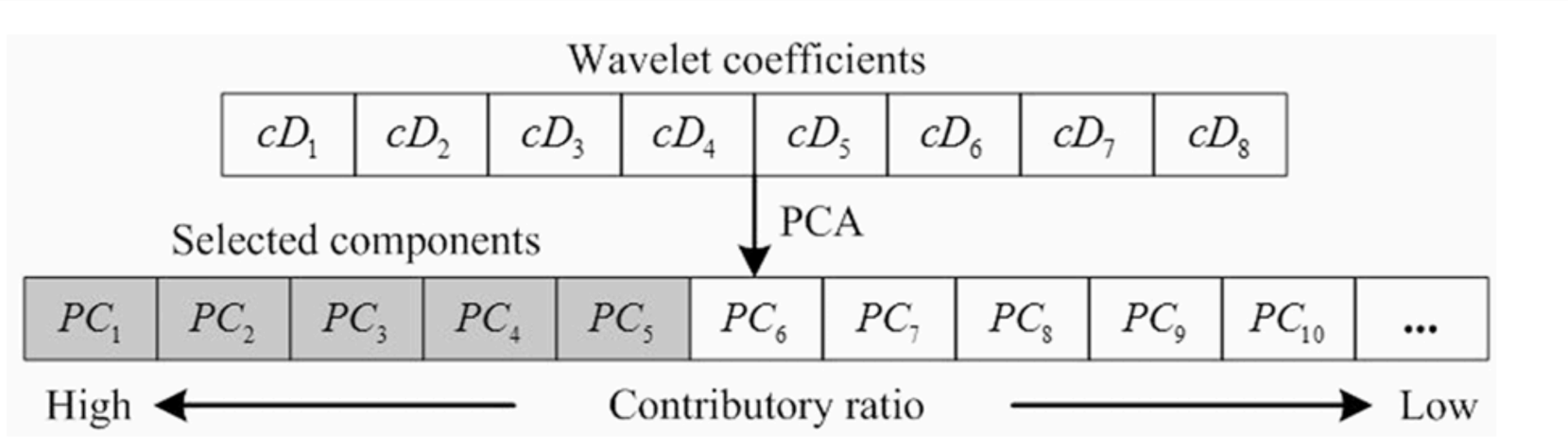


Figure 4. Low-dimensional feature vector generated by PCA using wavelet coefficients.

Source: Qin, Q. et al. Sci Rep 7, 6067 (2017)

Steps 4 : SVM Classification

- Classifier: one-vs-one Support Vector Machine with RBF kernel, C = 10, γ = 0.1
- Since there are 6 classes, 15 classifiers are used
- The 15 binary SVMs vote on the input beat, class recieving the majority of votes becomes the predicted class
- Evaluated with 10-fold cross-validation under two schemes: beat-based (random split) and record-based (patient-independent split)

Source: Qin, Q. et al. Sci Rep 7, 6067 (2017)

SECTION 04

Results

Evaluation protocol and metrics

Beat-base cross-validation	Record-base cross-validation
Individual beats distributed across 10 folds	Complete ECG records held out for testing
Approximately 90% training and 10% testing	Training and test records remain separate within each fold
Beats from the same record may occur in both sets	Measures generalization to unseen records

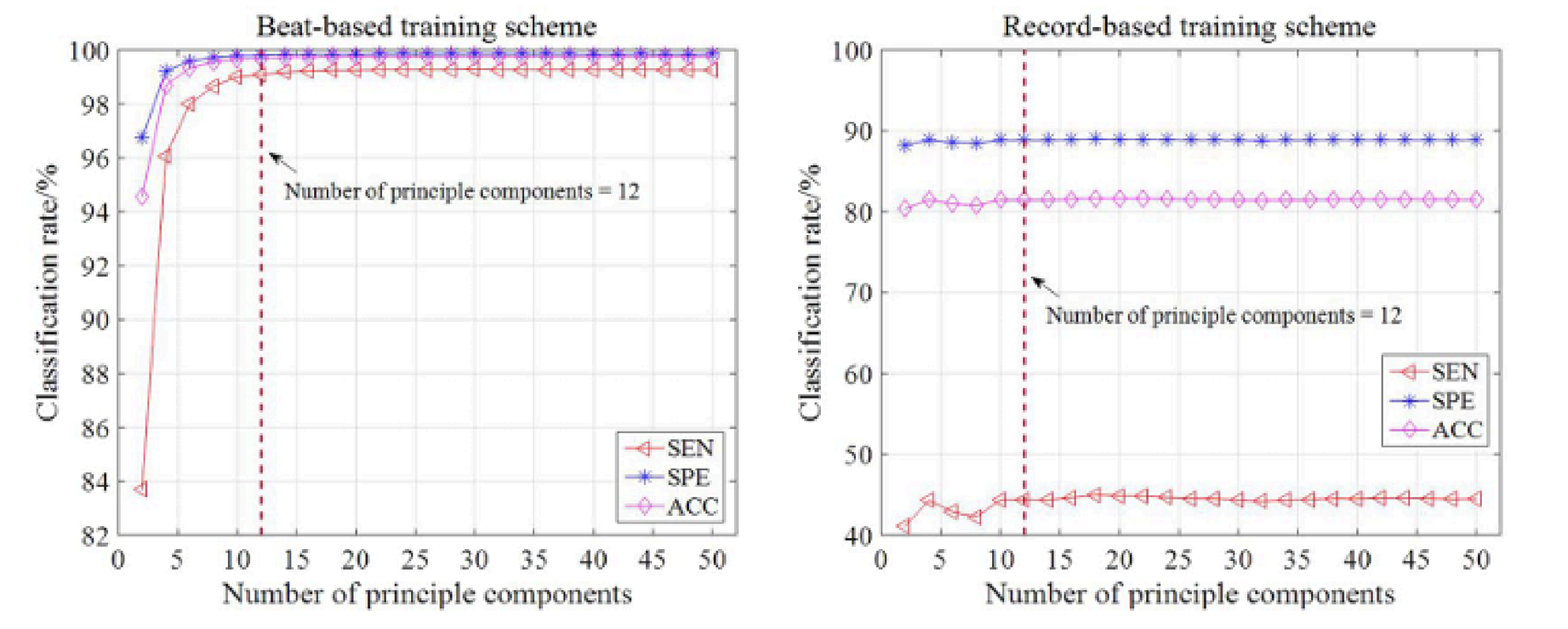
Metrics

$$SEN = \frac{TP}{TP + FN} \qquad SPE = \frac{TN}{TN + FP} \qquad ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

Scheme	Sensitivity	Specificity	Accuracy
Beat-based	99.09%	99.82%	99.70%
Record-based	44.40%	88.88%	81.47%

Source: Qin, Q. et al. Sci Rep 7, 6067 (2017)

Selection of 12 principal components

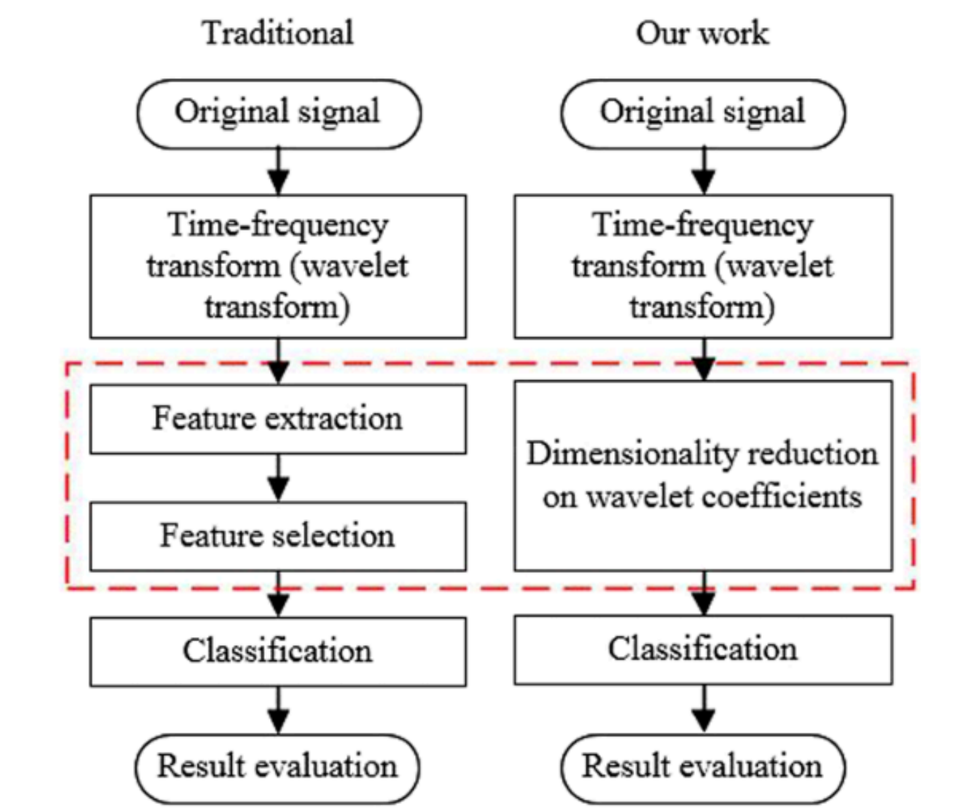


Classification rates with different numbers of principle components.

- Authors have evaluated principal components numbers from 2 to 50 evens.
- Observed behaviour :
  - Rapid initial performance improvement
  - Stable performance from approximately 12 components in both evaluation schemes. (3.2 % of the original dimension)

Source: Qin, Q. et al. Sci Rep 7, 6067 (2017), Table 5

Differences from previous work



Comparison between our method and traditional wavelet transform-based methods.

Feature / Aspect	Traditional Wavelet Approaches	This Study
Coefficient Selection	Selects coe. only from certain wd levels; separate features from different fre. bands	Uses all detail coefficients from cD1 to cD8
Feature Representation	Manual feature-selection methods	Concatenates into one 373-element vector
Dimensionality Reduction	Standard feature-subset selection	Applies PCA; retains 12 principal components
Dataset Scale	Smaller datasets or mixed-patient splits	Uses all 107,049 MIT-BIH beats
Validation Methodology	Unstandardized or mixed splits	Compares beat-based and record-based validation

Source: Qin, Q. et al. Sci Rep 7, 6067 (2017)

SECTION 05

Implementation Plan

Implementation Plan

STEP 1

Data preparation

- Download MIT-BIH ECG record and annotations.
- Select the lead 11 or MLII signal.
- Retain A, L, N, P, R and V annotations.

STEP 2

Feature extraction

- Extract 0.25 s before and 0.45 s after each R-peak
- Confirm 252 samples per beat
- Apply eight-level bior6.8 decomposition
- Concatenate cD1–cD8

STEP 3

Model pipeline

- Create beat-based and record-based folds
- Fit preprocessing and PCA using training data only
- Retain 12 PCA components
- Train RBF SVM with C=10 and gamma=0.1

STEP 4

Evaluation

- Reproduce sensitivity, specificity and accuracy.
- Add confusion matrix, precision, F1 and balanced accuracy.
- Compare the results.

Our Extension

Step 1

Problem : Class imbalance

- Use a weighted SVM.
- Tune class penalties using only training folds.
- Penalize mistakes on minority classes more strongly.

Step 2

Improved version : Morphological information + Temporal information

- Following the same procedure get the 12 PC scores.
- Make combined feature vector including 12 PCs + Previous + Next + Normalized RR interval lengths .
- Class-weighted one-versus-one RBF SVM.

Summary

- Dataset — MIT-BIH Arrhythmia Database, 107,049 beats across 6 classes
- Methods — beat segmentation → WMRA → PCA (12 components) → one-vs-one SVM
- Results — 99.70% accuracy (beat-based) vs. 81.47% accuracy (record-based)
- Implementation plan — 6 milestones, replicate then extend with ICA or VMD

Q & A

Thank you

Comparison between other related work and this proposed method

Literatures and feature extraction methods	Feature selection (dimension)	Beat types	Training/test beats	Classifiers	Independent training/test data	k-fold cross validation	*SEN* (%)	*SPE* (%)	*ACC* (%)
Spectral correlation[1]	Yes (88)	5	Totally 6259	SVM	Unknown	10-fold	99.20	99.70	98.60
Wavelet transform, morphological features[5]	No (28)	5	10675/93894	Artificial neural network	No	No	88.60	96.18	97.86
Morphological features[7]	Yes (6)	6	35848/35848	Linear discriminant analysis	No	No	91.19	98.65	94.03
Morphological features[8]	No (13)	3	600/30273	SVM, neural network	No	No	98.52	99.19	97.14
Time domain features[9]	No (9)	6	42427/14142	Decision tree	No	No	97.50	99.80	99.51
Morphological features[10]	No (16)	3	15509/8081	SVM, neural network	Yes	No	92.82	93.74	92.85
Morphological features[11]	No (8)	5	12570/12570	Regression neural network	No	No	85.50	99.40	99.40
Fourier transform, wavelet package[14]	Yes (70)	16	3345/2542	*k*-NN	No	No	85.59	99.56	93.59
Wavelet transform, cosine transform[15]	Yes (18)	4	720/360	SVM	Unknown	No	98.60	95.50	96.50
Wavelet transform[16]	Yes (24)	5	900/900	SVM, genetic algorithm	No	No	98.50	99.69	98.80
Higher order spectral[17]	No (7)	5	330/500	SVM	Unknown	No	90.00	87.93	85.79
Wavelet transform[18]	Yes (20)	4	360/360	SVM	Unknown	No	98.62	99.54	98.61
Temporal and spectral features[21]	Yes (15)	6	1440/720	SVM	No	No	97.60	93.80	95.20
Temporal and spectral features[22]	Yes (13)	8	Totally 17857	SVM	No	5-fold	95.00	99.00	98.60
Higher order statistics, wavelet packet[27]	Yes (28)	5	3345/2542	*k*-NN	Yes	No	89.80	97.80	—
Hilbert-Huang transform[32]	Yes (18)	6	10700/10700	SVM	No	No	98.64	99.77	99.51
Wavelet transform[46]	Yes (18)	5	Totally 101352	SVM	Yes	44-fold	—	—	86.40
		16	24100/86009		No	No	99.32	—	99.01
Approximate entropy, wavelet packet[47]	Yes (9)	5	145/145	SVM, PNN	Unknown	No	98.70	99.70	98.60
Non-linear and center-clipping transform[48]	No (5)	5	13640/13640	Wavelet neural network	No	No	98.78	99.70	98.78
Eigenvector method[49]	Yes (12)	4	360/360	Recurrent neural network	Unknown	No	98.89	99.25	98.06
Higher order statistics[50]	No (24)	5	4000/14299	RBF neural network	No	No	92.93	98.52	95.18
Geometrical features[51]	No (18)	7	4035/3150	SVM, *k*-NN, BPNN	No	No	97.52	99.65	98.06
Wavelet transform, morphological features[52]	Yes (8)	3	50928/49636	Linear discriminant analysis	Yes	No	80.00	—	94.00
Wavelet transform, linear prediction model[53]	No (12)	3	50554/49273	Linear discriminant analysis	Unknown	No	86.50	—	86.50
Cross correlation[54]	No (30)	3	41961/51285	Artificial neural network	Unknown	No	97.49	—	95.24
WMRA [This work]	**Yes (12)**	**6**	**Totally 107049**	**SVM**	**Yes**	**10-fold**	**44.40**	**88.88**	**81.47**
					No		**99.09**	**99.82**	**99.70**